

Electro Mechanical Systems (EE2015/2010)

Date: May 19th, 2026

Course Instructor(s)

Mr. Muhammad Abdul Majid

Mr Syed Ismael, Mr. Usama Faiz

Final Exam

Total Time: 3 Hours

Total Marks: 100

Total Questions: 04

Semester: SP-2026

Campus: Lahore

Dept: Electrical

Engineering

Azher Bashir
Student Name

24L6155 BEE4B
Roll No Section

[Signature]
Student Signature

Vetted by

Vetter Signature

Instructions:

Please **do not** ask any question, in case of an ambiguity write down your valid assumption and solve accordingly.

Show all the working and box out important answers.

CLO#2:

Analyze the performance and efficiency of practical single-phase and three-phase transformers using test data and equivalent circuit modeling

Q1:

[marks 25]

A 65-kVA, 4KV/0.23KV single phase distribution transformer has series impedance referred to the primary of $10 + j40 \Omega$. The components of the excitation branch referred to the secondary side are $R_C = 50 \Omega$ and $X_M = j60 \Omega$. Assume a secondary current of 209.09A with a lagging PF of 0.7273 is provided to a lagging load at rated secondary voltage. Analyze the performance of transformer and answer the following.

- a) Draw circuit model referred to secondary side and find out following

- b) Applied primary voltage
- c) Voltage regulation
- d) Real power input to the transformer
- e) Reactive power being delivered to the load

CLO #3:

Apply operating principles to model and solve performance- related problems of DC machines

Q2:

0.3 39V

[marks =25]

An 18KW, 250V compensated DC shunt motor has armature resistance of 0.3Ω . Its field resistance is 100Ω and adjustable resistance connected in series with field coil can vary from 55Ω to 257Ω . Its no load characteristic curve at a speed of 1000 rpm is tabulated as under.

0.1A $\rightarrow$ 13V

I_f (A)	0.3	0.6	0.9	1.2	1.4	1.5
E_a (V)	135	235	274	292	295	297

0.115 $\frac{39}{2}$ $\frac{15}{2} = \frac{39}{0.3}$

Apply the operating principles and model of DC machine to find out following quantities assuming rated terminal voltage in all cases.

- a) No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 127 ohm.
- b) When the motor is loaded it absorbs a line current of 61.2A, assuming same field current as in part (a), determine the speed at this loaded condition
- c) Power converted from electrical to mechanical in part b.
- d) Induced torque in part b
- e) Draw the torque speed curve of a non-compensated DC shunt motor

CLO #4:

Analyze the construction, operation, and performance characteristics of synchronous machines under various loading and excitation conditions

Q3:

synch gen

[marks 25]

An alternator is rated at 1250 KVA, 2.4 KV, 0.80-PF-lagging, 50-Hz, and has 28 pole. The machine is Y-connected and has a synchronous reactance of 1Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 28 KW, and its core losses are 24 KW. The field circuit has a dc voltage of 220 V, and the maximum I_f is 10 A. The resistance of the field circuit is adjustable over the range from 12 to 200 Ω .

The generator is providing full rated load at 0.85-PF-lagging, rated terminal voltage and 6A field current. **Analyze** the Synchronous Generator circuit to find out the following quantities

- a) Field circuit resistance
- b) Voltage regulation
- c) Efficiency
- d) Rotational speed, also comment about possible environmental impact of prime mover (strictly one line/sentence only)

- b) Applied primary voltage
- c) Voltage regulation
- d) Real power input to the transformer
- e) Reactive power being delivered to the load

CLO #3:

Apply operating principles to model and solve performance- related problems of DC machines

Q2:

[marks =25]

An 18KW, 250V compensated DC shunt motor has armature resistance of 0.3Ω . Its field resistance is 100Ω and adjustable resistance connected in series with field coil can vary from 55Ω to 257Ω . Its no load characteristic curve at a speed of 1000 rpm is tabulated as under.

I_f (A)	0.3	0.6	0.9	1.2	1.4	1.5
E_a (V)	135	235	274	292	295	297

Apply the operating principles and model of DC machine to find out following quantities assuming rated terminal voltage in all cases.

- a) No load speed of motor if adjustable resistance connected in series with field coil has been set to a value of 127 ohm.
- b) When the motor is loaded it absorbs a line current of 61.2A, assuming same field current as in part (a), determine the speed at this loaded condition
- c) Power converted from electrical to mechanical in part b.
- d) Induced torque in part b
- e) Draw the torque speed curve of a non-compensated DC shunt motor

CLO #4:

Analyze the construction, operation, and performance characteristics of synchronous machines under various loading and excitation conditions

Q3:

[marks 25]

An alternator is rated at 1250 KVA, 2.4 KV, 0.80-PF-lagging, 50-Hz, and has 28 pole. The machine is Y-connected and has a synchronous reactance of 1Ω and an armature resistance of 0.2Ω . At 50 Hz, its friction and windage losses are 28 KW, and its core losses are 24 KW. The field circuit has a dc voltage of 220 V, and the maximum I_f is 10 A. The resistance of the field circuit is adjustable over the range from 12 to 200 Ω .

The generator is providing full rated load at 0.85-PF-lagging, rated terminal voltage and 6A field current. **Analyze** the Synchronous Generator circuit to find out the following quantities

- a) Field circuit resistance
- b) Voltage regulation
- c) Efficiency
- d) Rotational speed, also comment about possible environmental impact of prime mover (strictly one line/sentence only)

- e) Also find out applied torque at the shaft of generator
-

CLO#5 Analyze the construction, operation and performance characteristics of induction machines under various starting conditions

Q4:

[marks 25]

A 35 HP, 60 Hz, 4 pole, Y connected Induction motor has following per phase parameters referred to stator side: $R_1 = 0.6 \Omega$, $R_2 = 0.3 \Omega$, $X_1 = 1 \Omega$, $X_2 = 0.4 \Omega$, and $X_M = 25 \Omega$. The mechanical and core losses are 600 W and 1000 W respectively. Rated terminal voltage is 519.62V. Analyze the operation and performance of the motor at rated voltage and frequency and at a slip of 3 % to find out following.

- (a) Stator current along with its angle
- (b) The air gap power
- (c) Load torque
- (d) Rotor copper loss
- (e) Also draw general torques speed characteristic curve of an Induction Motor